IOTAFLOW
SYSTEMS PVT. LTD

IOTAFLOW UF

ULTRASONIC CLAMP ON FLOW METER

"If You Want To Manage It, Measure It"

Advantages:

- Digital signal processing, high noise immunity;
- DN range from 10 mm up to 5000 mm;
- Stable operation in spite of weak signal on aged piping;
- Reliable metering on fluid parameters alteration;
- Easy commissioning;
- Bidirectional flow measurement;
- Clamp-on and wetted transducers;
- No pressure loss in pipeline;
- Batching mode to measure the dose;
- Multi-beam measuring;
- Up to 4 pipes with one electronic unit;
- Waterproof transducers;
- MID and OIML certified;
- ISO 9001:2008 certified.

Application:

- Hot and cold process and potable water
- Waste Water
- Acids
- Alkalis
- Liquid Foods

Model Details

UFM-1510-1021-XXXX-2-AO-H-IP65-1-S-M-BHEL

Ultrasonic Transit Time Clamp-on Flow Meter	
Series	1510
1021-XXXX	Probe Model and Series (Dn Min, Dn Max)
2	Dual Path (2 Channel)
AO	Isolated 4 - 20mA Output signal
H	Hart Communication enabled
IP65	Ingress Protection Ip65
1	220v AC Power Supply
S	ENCLOSURE SS316
M	Relay Output (2NO+2NC)

Optional Transducer

Types	Spec.	Model	Measurement Range
Clamp On	Small Size	1522	DN15~ DN100
	Medium Size	1523	DN50~ DN700
	Large Size	1524	DN300~ DN6000

Description and Operation

Application

IOTAFLOW "UF" 1500 series multichannel ultrasonic flowmeter is designed to measure average volumetric flow rate and volume of bidirectional flows of various liquids (hot, cold and waste waters, acids, caustics etc.) in one or several pressure pipelines under various operating conditions including explosion hazard areas. The flowmeter is suitable for a wide range of applications in power

industry, municipal engineering as well as in oil, gas, food and other industries. The flowmeters may be used as part of heat meters, metering systems, automatic process control systems etc.

IOTAFLOW "UF" 1500 series is suitable for measuring flow rate and volume of liquid food products including alcohol-free drinks (juices, syrups and the like), alcohol drinks, dairy products (milk, yoghurt, kefir, sour cream etc.), mayonnaise, ketchup, solutions of food acids and alkalis etc.

IOTAFLOW "UF" 1500 series multichannel ultrasonic flowmeter performs the following functions:
• Measuring average volumetric flow rate in either forward or reverse flow directions through 1-4 measuring channels (pipelines)
• Totalizing volume of forward and reverse flow independently and calculating their algebraic sum with regard to flow direction for each measuring channel
• Determining the current value of flow velocity and flow direction for each channel
• Batching preset liquid volume or batching in the start-stop mode along with determining the amount of batched liquid, batching time and flow rate
• Outputting measurement results to the analog (current), pulse or logical output
• Logging measurement data and configuration settings in the internal nonvolatile memory
• Displaying measurement, diagnostic, configuration and history (logged) data on the built-in indicator, data is transferred via RS-232 or RS-485 interface using dedicated cable, phone cable, Ethernet interface, radio channel or telecommunications channel
• Configuring flowmeter's settings according to on-site and process requirements- Monitoring and indicating alarm conditions (error situations) and faults; recording error and fault types and intervals into the dedicated logs - Providing protection of logged data and configuration settings from unauthorized access.

Specifications:

General Specifications are listed in Table

Table

Parameter	Value	Notes
1. Number of measuring channels	From 1 to 4	Optional
2. Nominal diameter of the pipeline, DN	From 50 to 5000	
3. Temperature of the medium, °C	From - 30 to +150	
4. Power supply voltage	DC 24 V	See section 1.2.6
5. Power consumption, W	no more than 12	
6. Mean time to failure, h	75 000	
7. Mean life time, years	12	

The flowmeter measures average volumetric flow rate up 20 m/s flow velocity on the base of the following formula:

Where Q – average volumetric flow rate, m³ /h;
v – flow velocity, m/s;
D – pipeline internal diameter, mm.

$$Q = 2.83.10^{-3}.v.D^2,$$

Volume is measured at a flow velocity value of no more than 10.6 m/s. This value can be increased up to 20 m/s. Flowmeter's sensitivity to flow velocity is no less than 0.01 m/s.

The flowmeter provides that the results of measurements can be outputted to:
• Universal outputs – from 1 to 9 (optional)
• Analog output – 2 (optional) - RS-232 (RS-485) interface – 1;
• Ethernet interface – 1 (optional).

The flowmeter provides storage of measurement results in the following logs: -

Hourly log – 1440 records (hourly records) 60 days
Daily log– 60 records (daily records)
Monthly log– 48 records (monthly records)
Interval log – up to 14400 records
Batcher log – up to 512 records per a channel
User log – up to 1000 records

Measuring channels error log- up to 512 records per a channel
Failure log- up to 60 records
Mode log- up to 512 records.
When powered off, the flowmeter stores measurement results and configuration settings for no less than 1 year
The flowmeter is powered from a stabilized 22 ... 29 V DC voltage source with pulse level of no more than ± 1.0%.
The flowmeter can be powered from mains AC (154-264) V (50±2) Hz via Secondary Power Source ADN-3024 (ADN).

Environmental restrictions:

a) Ambient temperature

- CS: ambient temperature from 0 to + 50 C (optional frost-resistant model designed for the temperature range of - 40 to + 65 C is available)
- Clamp-on PEAs: liquid temperature from - 30 to + 150 C

b) Relative humidity

- CS: up to 80 % under + 35 C, without moisture condensation
- PEAs: up to 100 % under maximum + 40 C, with moisture condensation.

c) Atmospheric pressure from 66.0 to 106.7 kPa;

Protection provided by the enclosure:

- CS: Ip54
- PEA: IP67

Metrological specifications

Limits of permissible relative error for measurement, indication, logging, storage and transferring of average volumetric flow rate and volume measurement results for any flow direction for flowmeters with primary converters1 (CPs) do not fall outside the following ranges and operation conditions comply with those specified in the present manual):

$\delta = \pm (0{,}95 + \frac{0{,}1}{v})$ – using single-beam sounding method with a liquid viscosity of ≤ 4 cSt;
$\delta = \pm (1{,}2 + \frac{0{,}2}{v})$ – using single-beam sounding method with a liquid viscosity of > 4 cSt;
$\delta = \pm (0{,}45 + \frac{0{,}1}{v})$ – using two-beam sounding method;
$\delta = \pm (0{,}25 + \frac{0{,}1}{v})$ – using four -beam sounding method.

Where δ – limits of permissible relative error, %; v – flow velocity, m/s.

NOTE. 1 “Primary converter” means a pipeline section with PEAs installed on it. The limits of permissible relative measurement error for measuring time of operation in various modes do not exceed 0.1 %.

Outputs

- LCD;
- Pulse or logic digital output (up to 8 additional digital outputs on request);
- RS-232/RS-485 serial interface (ModBus);
- 0-5, 0-20, or 4-20 mA analog output (on request);
- Ethernet (on request)

Model Decodification

UFM			
Ultrasonic Convertor	1510	Single Channel/Track (2 Nos. Sensors)	
	1520	Single Channel/Track (4 Nos. Sensors)	
	1540	Single Channel/Track (8 Nos. Sensors)	
Clamp-on Probes	1522	DN 15 - 100	(-30 to 90 Deg C)
	1523	DN 50 - 700	(-30 to 90 Deg C)
	1524	DN 300 - 6000	(-30 to 90 Deg C)
	V206	DN 100 - 3000 (Difficult Applications)	

Beyond the Best: Explore More

WATER FLOW METER

Ideal for measuring liquid flow with high accuracy and minimal pressure drop. The perfect choice for a wide range of industrial applications

POSITIVE DISPLACEMENT OIL METERS

Highly accurate for measuring viscous liquids. Perfect for industries like oil, chemical, and food processing where precise volume measurements are essential.

AIR FLOW METERS

Designed to monitor and control air flow in HVAC systems, compressed air systems, and more. Ensures efficiency and accurate flow measurement in air handling processes.

GAS METERS

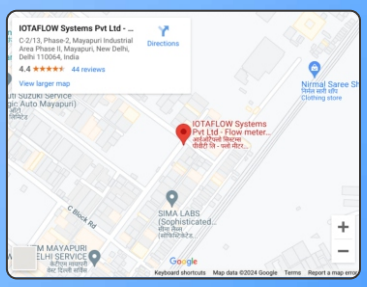
These meters ensure precise measurement of natural gas, biogas, and other gases, enhancing safety and operational efficiency.

Contact Us
IOTAFLOW SYSTEMS PVT. LTD.

C-2/13, Mayapuri Industrial Area Phase-II,
New Delhi - 110064 (INDIA)

+91-9910693866/ +91-97790 85987

contact@iotaflow.com



Specifications subject to change without prior notice. © Iotaflow Systems Pvt. Ltd.

www.iotaflow.com

Cat no. 103.02.0925